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A Comparative Study of Forecasting COVID Cases

Using Linear and Non-linear Forecasting Models

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1 Introduction

Daily COVID-19 cases are difficult to predict because of epidemic waves, reporting delays, and weekly reporting patterns. Accurate short-term forecasts are important because they help public health authorities monitor disease spread, allocate healthcare resources, and support in decision making. (Congdon, 2022; Refisch et al., 2022).

Time-series models are widely used for disease forecasting because they use past observations to predict future values. Autoregressive (AR) models are simple and interpretable, while autoregressive models with exogenous variables (ARX) include additional information such as weather, age , gender, and calendar effects. Machine-learning methods, including regression trees and random forests, can also model complex nonlinear relationships without requiring a predefined functional form. Past studies have shown that no single forecasting method performs best for all datasets, making empirical comparisons necessary (Aziz et al., 2023; Bergman & Fishman, 2023; Bosse et al., 2022; Canino et al., 2022; Gao et al., 2021; Rajab et al., 2022; Zhang et al., 2022).

Germany provides an interesting case study because the COVID data published by the Robert Koch Institute contain multiple epidemic waves, reporting delays, and strong weekly patterns. This study compares one-step-ahead forecasts of daily reported COVID-19 cases using four forecasting methods: an autoregressive (AR) model, an autoregressive model with exogenous variables (ARX), a regression tree, and a random forest. Forecasts are generated using a recursive forecasting framework with training, validation, and test samples. Forecast accuracy is evaluated using the Root mean squared forecast error (RMSFE), while the Diebold–Mariano test is used to assess whether differences in forecasting performance are statistically significant (Refisch et al., 2022).

Four predictor specifications are applied consistently across all forecasting methods. The benchmark model uses only lagged COVID-19 cases, while the remaining specifications includes covariates like temperature , start of the week ,.etc. In addition, Seasonal Autoregressive Models with Exogenous Variables (SARX) are estimated to examine whether annual seasonal patterns improve forecast accuracy. (Peng et al., 2025; Zhang et al., 2022).

Model selection for the autoregressive models is based on the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC), which provide different trade-offs between model fit and complexity (Akaike, 1974; Ding et al., 2018; Schwarz, 1978; Vrieze, 2012).

The remainder of the paper is organised as follows. Section 2 describes the data and forecasting design. Section 3 presents the forecasting methods and model specifications. Section 4 reports the empirical results and statistical tests. Section 5 presents the SARX analysis. Section 6 concludes.

2 Problem and Data Description

2.1 Research Objectives

This study compares the out-of-sample forecasting performance of statistical and machine-learning models for daily reported COVID-19 cases in Germany. Forecasts are generated and evaluated one step ahead on a common test sample, with accuracy measured by the the Root mean squared forecast error (RMSFE).

Another objective is to examine whether temperature, age-group shares, gender shares, calendar effects, and reporting-related variables improve forecast accuracy relative to models based solely on lagged case counts. The models considered are the autoregressive model (AR), the autoregressive model with exogenous variables (ARX), the seasonal autoregressive model (SAR), the regression tree (RT), and the random forest (RF). Forecast accuracy is compared using the Diebold–Mariano test.

Research Hypotheses

The empirical analysis tests the following hypotheses:

H1 Models including additional variables produce lower out-of-sample forecast errors than models based only on lagged case counts.

H2 Random forest and regression tree models achieve lower out-of-sample forecast errors than the linear ARX models.

H3 The differences in forecast accuracy between forecasting models are statistically significant according to the Diebold–Mariano test.

2.2 Data Sources

The primary data source is the publicly available *SARS-CoV-2 Infections in Germany* dataset provided by the Robert Koch Institute (RKI)(Robert Koch Institute, 2026).The RKI dataset provides nationwide information on confirmed COVID-19 cases reported by local health authorities. As the dataset is continuously updated when new information becomes available or existing records are corrected, the reported case counts should be interpreted as dynamic administrative records rather than fixed historical observations.

Daily average temperature data were obtained from the German Meteorological Service (Deutscher Wetterdienst, DWD(Deutscher Wetterdienst (DWD), 2025)). These observations were merged with the COVID-19 data by reporting date (*Meldedatum*) and used as a temperature predictor.

2.3 Data Structure

The original RKI dataset contains approximately 7.8 million individual records of confirmed COVID-19 cases. The analysis is restricted to observations reported between 1 April 2020 and 31 March 2026. This period covers several phases of the pandemic and provides sufficient observations for model estimation and out-of-sample evaluation.

The original records are reported at district level. Since the objective is to forecast national daily case counts, the data were aggregated by reporting date (*Meldedatum*).Each observation represents the total number of reported COVID-19 cases in Germany on that date. The resulting series contains 2,191 daily observations before the construction of lagged variables.

2.4 Variables

The forecasting models use the daily number of reported COVID-19 cases as the target variable. The explanatory variables are daily average temperature, age-group shares, gender shares, reporting delay, symptom-onset share, a public holiday indicator, and a start-of-week indicator. A detailed description of the variables available in the original RKI dataset is provided in Appendix A.

2.4.1 Target Variable

The target variable is the daily number of reported COVID-19 cases in Germany. It is constructed by aggregating the district-level case counts (*AnzahlFall*) according to the reporting date (*Meldedatum*). To stabilise the variance, the target variable is transformed as

$$y_t^* = \log(y_t + 1), \tag{1}$$

where y_t denotes the observed number of reported cases on day t. The constant one ensures that the transformation is defined for zero case counts. All models are estimated using the transformed series, while forecasts are converted back to the original scale for evaluation and interpretation.

2.4.2 Covariates

Past values of the covariates are used except for variables that are known in advance, such as start of the week. This ensures that no future information is used to generate the forecasts. The following variables are used as covariates in forecasting.

Temperature. Let $T_{i,t}$ denote the daily mean temperature (TMK) recorded at weather station i on day t. The national daily average temperature is computed as

$$Temp_t = \frac{1}{N_t} \sum_{i=1}^{N_t} T_{i,t},$$

where N_t is the number of weather stations with valid observations on day t. To allow for delayed weather effects, lagged temperature variables are included in the forecasting models,

$$Temp_{t-l}, \qquad l = 1, \ldots, 10,$$

where $Temp_{t-l}$ denotes the average temperature observed l days before day t.

Age and Gender variables. The daily age-group and gender shares are defined as

$$AgeShare_{g,t} = \frac{Cases_{g,t}}{\sum_g Cases_{g,t}},$$

and

$$GenderShare_{s,t} = \frac{Cases_{s,t}}{\sum_s Cases_{s,t}},$$

where $Cases_{g,t}$ and $Cases_{s,t}$ denote the number of reported cases in age group g and gender category s on day t, respectively.

Calendar variables. Weekly reporting effects are represented by a start-of-week indicator,

$$StartOfWeek_t = \begin{cases} 1, & \text{if day } t \text{ is Monday,} \\ 0, & \text{otherwise,} \end{cases}$$

and public holidays are represented by

$$Holiday_t = \begin{cases} 1, & \text{if day } t \text{ is a public holiday,} \\ 0, & \text{otherwise.} \end{cases}$$

The seventh lag, $Holiday_{t-7}$, is used in the forecasting models because daily COVID-19 reports follow a strong weekly reporting cycle. A lag of seven compares the current day with the same day of the previous week and can capture delayed reporting or backlog effects associated with public holidays. (Refisch et al., 2022)

Reporting delay. Reporting delay is defined as the difference between the reporting date and the symptom-onset date,

$$Delay_{i,t} = Meldedatum_{i,t} - Refdatum_{i,t},$$

for records with *IstErkrankungsbeginn*=1. To reduce the influence of extreme observations, the delay is restricted to the interval $[-30, 60]$:

$$Delay^*_{i,t} = \min\left\{\max\left(Delay_{i,t}, -30\right), 60\right\}.$$

The daily average reporting delay is

$$ReportingDelay_t = \frac{1}{N_t}\sum_{i=1}^{N_t} Delay^*_{i,t},$$

where N_t denotes the number of records with usable symptom-onset information on day t.

Symptom-onset share. The symptom-onset share is defined as

$$SymptomOnsetShare_t = \frac{OnsetCases_t}{TotalNewCases_t},$$

where

$$OnsetCases_t = \sum_i AnzahlFall_{i,t} \cdot I\left(NeuerFall_{i,t} = 1\right) \cdot I\left(IstErkrankungsbeginn_{i,t} = 1\right),$$

and

$$TotalNewCases_t = \sum_i AnzahlFall_{i,t} \cdot I\left(NeuerFall_{i,t} = 1\right).$$

The measure is based only on newly reported cases($NeuerFall_{i,t} = 1$)because historical cases are repeatedly included in RKI data. Restricting the calculation to new cases provides a more accurate measure of reporting and makes the lagged predictor more informative.

2.5 Data Quality and Limitations

The RKI dataset provides nationwide surveillance data for COVID-19 cases in Germany. The reported values can change over time because local health authorities may correct existing records or add new information after the initial release. Therefore, the dataset should

be treated as dynamic rather than fixed. Some observations also contain unknown age, gender, or symptom-onset information, which affects the construction of some covariates.

Reported COVID cases are also influenced by the reporting process. Weekly reporting patterns and public holidays can change the number of cases reported on a given day. These effects are captured by the calendar variables included in the forecasting models. The analysis focuses on forecasting daily reported COVID-19 cases for Germany rather than regional forecasts.

2.6 Training, Validation and Test Sets

After constructing the lagged variables, the dataset contains 2,181 daily observations. The observations are divided into a training set of 1,395 observations, a validation set of 349 observations, and a test set of 437 observations. The training set is used for model estimation, the validation set for selecting lag structures and hyperparameters, and the test set for the final out-of-sample evaluation.

3 Methodology

3.1 Software and Computational Environment

The empirical analysis was conducted using the statistical software R (R Core Team, 2025). Linear autoregressive models were estimated using functions from the `stats` package (R Core Team, 2025). Regression tree models were estimated using the `rpart` package (Therneau & Atkinson, 2023). Random forest models were implemented using the `ranger` package (Wright & Ziegler, 2017). Forecast accuracy was evaluated using the Root Mean Squared Forecast Error (RMSFE), which was computed directly in R. The Diebold–Mariano test was conducted using the `forecast` package (Hyndman & Athanasopoulos, 2025).

3.2 General Forecasting Framework

The objective of the empirical analysis is to generate one-step-ahead forecasts of the daily number of reported COVID-19 cases. The general forecasting problem is written as

$$y_t = f(y_{t-1}, \ldots, y_{t-p}, \mathbf{X}_{t-1}, \ldots, \mathbf{X}_{t-p}) + \varepsilon_t, \tag{2}$$

where y_t denotes the log-transformed number of reported COVID-19 cases on day t, $\mathbf{X}_t$ represents the vector of exogenous covariates, p is the maximum lag order, and ε_t is an unpredictable disturbance term. The function $f(\cdot)$ may be linear or nonlinear depending on the forecasting method.

3.2.1 Lag-Length Selection Using AIC and BIC

The number of autoregressive lags determines how much historical information enters the model. Too few lags may leave serial dependence in the errors, while too many lags increase estimation uncertainty and may reduce out-of-sample accuracy. lag lengths are therefore compared using the Akaike Information Criterion and the Bayesian Information Criterion (Akaike, 1974; Schwarz, 1978).

For a model with maximized likelihood $\widehat{L}$, k estimated parameters, and n observations, the criteria are

$$AIC = -2\log(\widehat{L}) + 2k, \tag{3}$$

$$BIC = -2\log(\widehat{L}) + k\log(n). \tag{4}$$

The preferred lag order is the value that minimizes the selected criterion. AIC usually gives a smaller penalty for additional lags and is often useful when forecast accuracy is the main objective. BIC applies a stronger penalty and tends to select more specifications. In this report both criteria are reported. When they select different orders, the final choice is checked using validation-sample forecast accuracy rather than relying only on in-sample fit (Lütkepohl, 2005; Shibata, 1980).

3.2.2 Model Specifications

To ensure a fair comparison across forecasting methods, the same predictor specifications are used for the ARX, SARX, regression tree, and random forest models. Each specification combines lagged COVID-19 cases with different groups of exogenous variables.

The following four model specifications are considered throughout the empirical analysis:

- **Lagged cases only:** Lagged daily reported COVID-19 cases.
- **Model 1:** Lagged cases, temperature, age-group shares, gender shares, start-of-week indicator, and public-holiday indicator.
- **Model 2:** Lagged cases, start-of-week indicator, public-holiday indicator, reporting delay, and symptom-onset share.
- **All covariates:** Lagged cases together with temperature, age-group shares, gender shares, start-of-week indicator, public-holiday indicator, reporting delay, and symptom-onset share.

The lagged-cases model serves as the benchmark. The remaining models are used to evaluate whether exogenous variables improve forecast accuracy. The same predictor variables are used in all forecasting methods to ensure that differences in forecasting

performance are caused by the modelling approach and not by differences in the available information.

3.2.3 One-Step-Ahead Forecasting

The forecasting exercise is based on one-step-ahead predictions. At forecast origin t, the model uses information available up to day t to predict the observation on day $t+1$.(Hyndman & Athanasopoulos, 2021; West & Harrison, 1996).At each forecast origin, only information that would have been available on that day is used to predict the following observation. The one-step-ahead forecast is therefore given by

$$\widehat{y}_{t+1} = \widehat{f}_t(y_t, \ldots, y_{t-p+1}, \mathbf{X}_t, \ldots, \mathbf{X}_{t-p+1}) \,, \tag{5}$$

where $\widehat{f}_t(\cdot)$ denotes the forecasting model estimated using the information available up to time t. For each forecast date, all lags and predictors must be observable at the corresponding forecast origin.

3.3 Training Windows

3.3.1 Expanding-Window Forecasts

An expanding window begins with an initial training sample and adds one new observation after every forecast. Earlier observations are never removed. Suppose the first estimation window ends at T_0. The model used to forecast y_{T_0+h} is estimated using observations from the beginning of the sample through $T_0 + h - 1$ (Tashman, 2000; West & Harrison, 1996).

For the hth one-step-ahead forecast, the estimation set is

$$\mathcal{W}_h^E = \{1, 2, \ldots, T_0 + h - 1\}. \tag{6}$$

The expanding-window approach uses all available historical observations to estimate the model. It is appropriate when the relationship between the target variable and the predictors is stable over time. The expanding-window results are compared with the rolling-window results to check the robustness of the findings (Clark & McCracken, 2001; Giacomini & White, 2006).

3.3.2 Rolling-Window Forecasts

A rolling window uses a fixed number of the most recent observations. After each forecast, the oldest observation is removed and the newest observation is added. Let $\mathcal{W}$ denote the window length. For forecast origin t, the estimation sample is

$$\mathcal{W}_t^R = \{t - W + 1, \ldots, t\}. \tag{7}$$

This approach allows the model coefficients to change over time as new data become available. This is useful for COVID-19 forecasting because the relationship between cases, calendar effects, and reporting practices can change during the pandemic. (Giacomini & White, 2006; Pesaran et al., 2013).

The rolling-window evaluation also ensures that future observations are not used when generating forecasts. All data transformations, lag selection, and parameter tuning are carried out using only the training and validation data available at that time. This makes the forecasting exercise similar to a real-time forecasting setting (Bergmeir et al., 2018; Tashman, 2000).

3.4 Linear Forecasting Models

3.4.1 Autoregressive Models

The autoregressive model is used as the main statistical benchmark. It explains the current transformed value using its own previous values. This is suitable for COVID-19 cases because daily observations show persistence and repeated weekly reporting patterns (Box et al., 2015; Hamilton, 1994).

An autoregressive model of order p, denoted AR(p), is

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t, \tag{8}$$

where c is a constant, ϕ_i are autoregressive coefficients, and ε_t is an error term with zero conditional mean. The AR model measures how much of tomorrow's reported value can be predicted from the recent history of the same series. It supplies a transparent benchmark against which the additional value of covariates and nonlinear models can be evaluated (Hyndman & Athanasopoulos, 2021; Lütkepohl, 2005).

Autoregressive Models with Exogenous Variables

The Autoregressive Models with Exogenous Variables(ARX) model extends the autoregressive benchmark by adding predictors that may contain information not already captured by past values of the target. The model is written as

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \boldsymbol{\beta}' \mathbf{x}_t + \varepsilon_t, \tag{9}$$

where $\mathbf{x}_t$ is a vector of predictors and $\boldsymbol{\beta}$ contains their coefficients. (Hamilton, 1994; Hyndman & Athanasopoulos, 2021).

The predictor variables include temperature, weekday and public holiday indicators, age-group and gender shares, and lagged reporting-related variables.

3.4.2 Linear Predictive Models Estimated by OLS

The benchmark forecasting models are estimated using Ordinary Least Squares (OLS). Let y_t^* denote the transformed daily COVID-19 case count and let $\mathbf{x}_t$ denote the vector of predictor variables, which includes lagged case counts and the selected covariates. The linear forecasting model is

$$y_t = \beta_0 + \mathbf{x}_t'\boldsymbol{\beta} + \varepsilon_t, \tag{10}$$

where β_0 is the intercept, $\boldsymbol{\beta}$ is the vector of regression coefficients, and ε_t is the error term.

The regression coefficients are estimated by Ordinary Least Squares, which minimises the sum of squared residuals,

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{t=1}^{T} \left(y_t - \beta_0 - \mathbf{x}_t'\boldsymbol{\beta}\right)^2. \tag{11}$$

The predictor vector $\mathbf{x}_t$ differs across the four model specifications considered in this study. (Hyndman & Athanasopoulos, 2021; Wooldridge, 2016).

3.5 Non-Linear Forecasting Models

3.6 Regression Trees

Regression trees are non-parametric forecasting models that divide the predictor space into smaller regions. Unlike linear regression models, they do not assume a linear relationship between the predictors and the response variable. Instead, they identify the predictor variables and split points that best separate the data into groups with similar response values. This allows regression trees to capture nonlinear relationships and interactions between predictors (Breiman et al., 1984; Hastie et al., 2009).

Let the predictor vector be

$$\mathbf{x}_t = (x_{1t}, x_{2t}, \ldots, x_{Kt})'.$$

For a predictor variable x_j and a split point s, the data are divided into two regions,

$$R_1(j,s) = \{\mathbf{x} : x_j \leq s\}, \qquad R_2(j,s) = \{\mathbf{x} : x_j > s\}.$$

For every possible split, the algorithm calculates the within-region residual sum of squares (RSS) and selects the split that minimises

$$RSS(j,s) = \sum_{\mathbf{x}_i \in R_1(j,s)} (y_i - \bar{y}_{R_1})^2 + \sum_{\mathbf{x}_i \in R_2(j,s)} (y_i - \bar{y}_{R_2})^2 . \tag{12}$$

The algorithm chooses the predictor variable and split point that minimise the residual sum of squares,

$$(j^*, s^*) = \arg\min_{j,s} RSS(j,s). \tag{13}$$

where $\bar{y}_{R_1}$ and $\bar{y}_{R_2}$ are the average values of the response variable in the two regions. The same splitting procedure is repeated until a stopping rule is reached. For a new observation, the forecast is the average value of the training observations in the terminal node to which the observation belongs.

To avoid overfitting, the size of the tree is controlled using three tuning parameters: the minimum number of observations required for a split, the maximum tree depth, and the cost-complexity parameter (*cp*). These tuning parameters are selected using the validation sample by choosing the values that produce the lowest Root Mean Squared Forecast Error (RMSFE) (Breiman et al., 1984; Kuhn & Johnson, 2013).

The four predictor specifications described in Section 3.2.2 are applied to the regression tree model using the recursive training windows described in Section 3.3.

3.7 Random Forests

Random forests are an extension of regression trees. Instead of using a single tree, the method combines the predictions from many regression trees to produce a single forecast. This reduces the variability of individual trees and generally improves forecasting accuracy (Biau & Scornet, 2016; Breiman, 2001).

The algorithm first draws B bootstrap samples from the training data. A regression tree is fitted to each bootstrap sample. At every split, only a random subset of the available predictor variables is considered when selecting the best split. This introduces diversity among the trees and reduces the correlation between them.

Let $T_b(\mathbf{x})$ denote the prediction from tree b for the predictor vector $\mathbf{x}$. The random forest forecast is obtained by averaging the predictions from all trees,

$$RF(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}), \tag{14}$$

The final forecast is therefore the average prediction across all trees. Averaging the

predictions reduces the effect of individual trees that may overfit the training data while preserving the ability to capture nonlinear relationships and interactions among the predictor variables (Breiman, 2001; Hastie et al., 2009).

The main tuning parameters are the number of trees (B), the number of predictor variables considered at each split (`mtry`), and the minimum terminal-node size. These tuning parameters are selected using the validation sample by choosing the combination that gives the lowest Root Mean Squared Forecast Error (RMSFE).

The four predictor specifications described in Section 3.2.2 are applied to the random forest model using the recursive training windows described in Section 3.3.

3.7.1 Hyperparameter Tuning

Model hyperparameters are selected using the validation sample. For each specification, one-step-ahead forecasts are generated on the validation period and the Root Mean Squared Forecast Error (RMSFE) is computed.(Bergmeir et al., 2018; Cerqueira et al., 2020; Kuhn & Johnson, 2013; Tashman, 2000) The optimal hyperparameter vector is chosen as

$$\widehat{\lambda} = \arg\min_{\lambda \in \Lambda} RMSFE_{val}(\lambda), \tag{15}$$

where Λ denotes the set of candidate hyperparameter combinations.

The tuning grids used for the regression tree and random forest models are reported in Table 1.

Table 1: Hyperparameter grids for the machine-learning models

Method	**Hyperparameter**	**Candidate values**
Regression Tree	Minimum split size	10, 20, 30, 50
	Maximum tree depth	3, 5, 7, 10
	Complexity parameter (cp)	0.001, 0.005, 0.010, 0.020
Random Forest	Number of trees	100, 200, 300
	Minimum node size	1, 5, 10, 20
	$mtry$	Determined by the number of predictors in each specification

3.8 Seasonal Autoregressive Models

Seasonal autoregressive (SAR) models account for both the short-term dependence between consecutive observations and the seasonal dependence that repeats every s periods (Box et al., 2015; Hyndman & Athanasopoulos, 2021).

A seasonal autoregressive model is written as

$$\Phi(B^s)\phi(B)y_t = c + \varepsilon_t, \tag{16}$$

where B is the lag operator, $\phi(B)$ represents the ordinary autoregressive terms, and $\Phi(B^s)$ represents the seasonal autoregressive terms. In this study, the seasonal period is set to $s = 7$ to capture the weekly reporting pattern in the daily COVID-19 data (Ghysels & Osborn, 2001; Hamilton, 1994).

Seasonal Autoregressive Models with Exogenous Variables

The Seasonal Autoregressive Model with Exogenous Variables (SARX) extends the seasonal autoregressive model by including additional predictor variables. The model combines short-term autoregressive effects, seasonal dependence, and exogenous variables in a single equation,

$$y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{P} \Phi_j y_{t-js} + \boldsymbol{\beta}'\mathbf{x}_t + \varepsilon_t. \tag{17}$$

The SARX model is used to examine whether the exogenous variables improve forecasting performance after accounting for both short-term dependence and the weekly seasonal pattern in the COVID-19 case data (Box et al., 2015; A. C. Harvey, 1990).

3.9 Root Mean Squared Forecast Error

Forecast accuracy is evaluated using the Root mean squared forecast error. Let $e_t = y_t - \widehat{y}_{t|t-1}$ denote the one-step-ahead forecast error and let N be the number of forecasts in the evaluation sample. RMSFE is

$$RMSFE = \sqrt{\frac{1}{N}\sum_{t=1}^{N} e_t^2}. \tag{18}$$

A lower RMSFE indicates better point-forecast accuracy. Squaring the errors gives a larger penalty to large mistakes. (Gneiting, 2011; Hyndman & Koehler, 2006).

3.10 Diebold–Mariano Test

Differences in the Root Mean Squared Forecast Error (RMSFE) do not necessarily imply that one forecasting model is better than another. Some differences may occur because of sampling variation. Therefore, the Diebold–Mariano (DM) test is used to compare the predictive accuracy of two competing forecasting models (Diebold & Mariano, 1995; D. Harvey et al., 1997).

Let $e_{1,t}$ and $e_{2,t}$ denote the forecast errors from Models 1 and 2. Under the squared-error loss function, the loss differential is

$$d_t = e_{1,t}^2 - e_{2,t}^2. \tag{19}$$

The null hypothesis is

$$H_0 : E(d_t) = 0, \tag{20}$$

which states that the two models have the same forecasting accuracy. A negative value of the average loss differential indicates that Model 1 has a lower forecast error, while a positive value favours Model 2.

The Diebold–Mariano test statistic is

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\mathrm{Var}}(\bar{d})}}, \tag{21}$$

where $\bar{d}$ is the sample mean of the loss differential and $\widehat{\mathrm{Var}}(\bar{d})$ is a consistent estimate of its variance.

Under the null hypothesis,

$$DM \overset{a}{\sim} N(0,1), \tag{22}$$

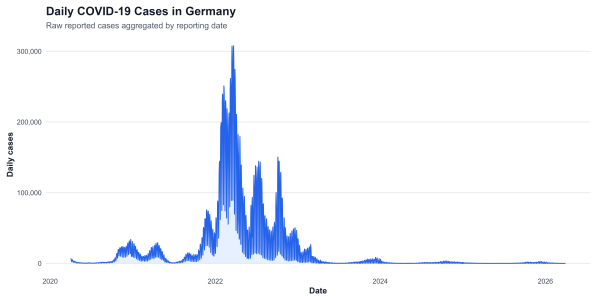
where $\overset{a}{\sim}$ denotes asymptotic convergence in distribution.

4 Empirical Analysis

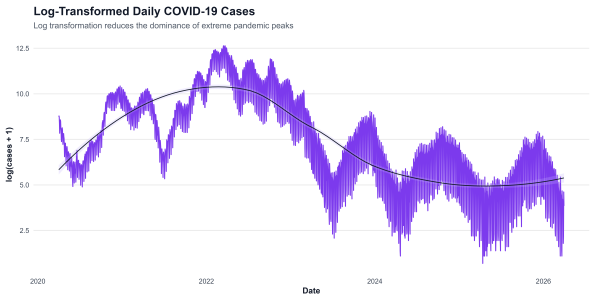
4.1 Descriptive Time-Series Evidence

Figure 1 presents the daily reported COVID-19 case series on the original and logarithmic scales. Panel 1a shows that the variation in daily cases rises during peak periods, indicating that the variance depends on the level of the series. Panel 1b shows that the transformation $\log(y_t + 1)$ reduces the influence of the largest peaks while preserving the cases patterns. The transformed series is therefore used for model estimation.

Figure 2 presents the sample autocorrelation and partial autocorrelation functions for the log-transformed series in the initial training sample. The autocorrelations remain large over the reported lags and display recurring peaks at multiples of seven, indicating strong persistence and a weekly reporting pattern. The partial autocorrelation function shows a dominant first-order effect and additional spikes at seasonal lags, particularly around lag seven.

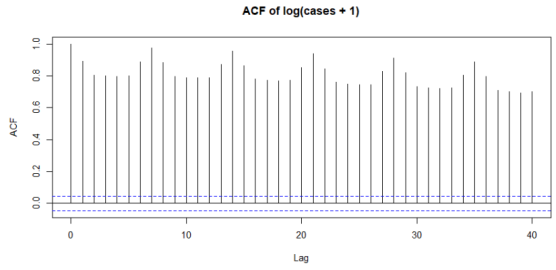


(a) Daily cases on the original scale.

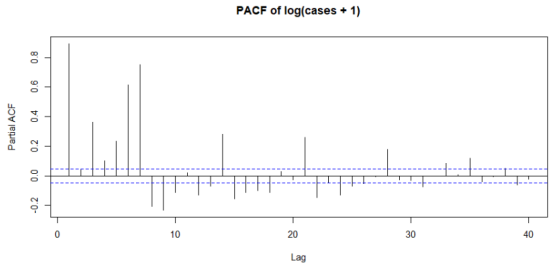


(b) Daily cases on the logarithmic scale.

Figure 1: Daily reported COVID-19 cases in Germany.



(a) Sample autocorrelation function.



(b) Sample partial autocorrelation function.

Figure 2: ACF and PACF of the log-transformed daily case series in the initial training sample.

4.2 Lag-Order Selection by AIC and BIC

The autoregressive lag order was selected using the initial training sample by estimating AR models with lag orders from $p = 0$ to $p = 10$. Model selection was based on the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). The results are reported in Table 2.

Table 2: AIC and BIC values for autoregressive models of order 0–10

Lag order (p)	AIC	BIC
0	7915.040	7925.968
1	5120.345	5136.737
2	5120.042	5141.898
3	4871.119	4898.438
4	4856.392	4889.176
5	4758.241	4796.489
6	3914.850	3958.562
7	2333.964	2383.139
8	2276.193	2330.832
9	2188.311	2248.415
10	**2169.048**	**2234.615**

Both information criteria decrease substantially as additional lags are included, indicating

that higher-order autoregressive models provide a better representation of the serial dependence in the data. Among the specifications, the AR(10) model achieves the lowest AIC and BIC values and is therefore selected as the preferred autoregressive specification. This model is subsequently used as the baseline autoregressive component in the forecasting models.

4.3 Validation of the Covariate Lag Structure

Before estimating the forecasting models, the lag structure of the exogenous variables was selected using the validation sample. The objective was to determine how much historical information from the explanatory variables should be included to improve short-term forecast accuracy. Four alternative ARX specifications were considered, in which the covariates entered either through lag 7 only, lag 10 only, lags 1–7, or lags 1–10. The autoregressive component was kept unchanged so that differences in forecast performance reflected only the choice of covariate lags. For this validation check , Model 1 was used.

Table 3: Validation results for linear ARX models with alternative covariate lag structures

Model	RMSFE (Original Cases)
ARX: Lag 7 only	170.34
ARX: Lag 10 only	177.67
ARX: Lags 1–10	226.77
ARX: Lags 1–7	241.99

Using only the seventh lag of the covariates produces the lowest RMSFE on the original case scale (170.34 cases). The lag-7 specification captures the weekly reporting cycle identified in the time-series analysis by the validation results.So, The lag-7 structure was used in the estimation of the ARX and related forecasting models.

4.4 Benchmark Autoregressive Models

The benchmark models are the autoregressive models with seven and ten lags, denoted as AR(7) and AR(10). These models use only lagged values of the daily COVID-19 cases to generate forecasts. The AR(7) model includes one complete weekly reporting cycle, while the AR(10) model includes three additional lags to capture more short-term information.

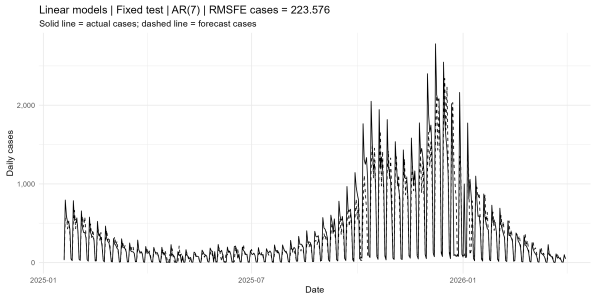
Table 4 reports the forecasting performance of the benchmark models for the fixed-test, expanding-window, and rolling-window forecasting designs.

Table 4: Out-of-sample performance of the benchmark autoregressive models

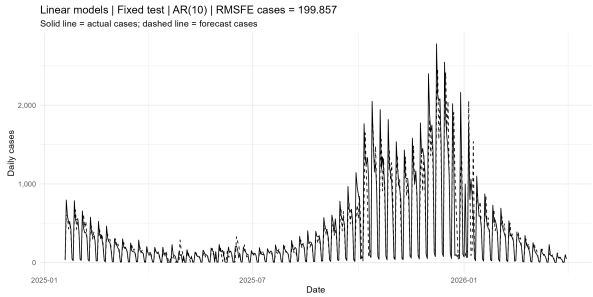
Model	**Forecast design**	**RMSFE (Cases)**
AR(10)	Fixed test	**199.76**
AR(10)	Expanding window	202.05
AR(10)	Rolling window	202.55
AR(7)	Fixed test	223.19
AR(7)	Expanding window	217.49
AR(7)	Rolling window	215.45

The results show that the AR(10) model performs better than the AR(7) model under all three forecasting designs. The lowest forecast error is obtained by the AR(10) model using the fixed-test design, with an RMSFE of 199.76 cases. For the AR(7) model, the lowest RMSFE is 215.45 cases under the rolling-window design. These results suggest that including additional lagged values improves forecasting accuracy.

Figures 3a and 3b compare the observed daily COVID-19 cases with the one-step-ahead forecasts from the AR(7) and AR(10) models. Both models follow the overall pattern of the data and capture the persistence in daily case numbers. However, larger forecast errors are observed during periods when case numbers change rapidly, especially during major pandemic waves. Since these models use only past case values, they do not account for other factors such as weather conditions, demographic variables, reporting behaviour, or other external information that may affect future case numbers.



(a) Observed and predicted cases using the AR(7) model.



(b) Observed and predicted cases using the AR(10) model.

Figure 3: Observed and one-step-ahead forecasts from the benchmark autoregressive models.

The forecasting results are similar across the three forecasting designs. The expanding-window and rolling-window models produce RMSFE values close to those obtained from the fixed-test design. This suggests that the relationship between current and past COVID-19 cases remains fairly stable during the evaluation period.

4.5 Linear Forecasting Models

The linear models include the AR benchmarks and ARX specifications with alternative covariate sets. Table 5 reports the lowest test-sample RMSFE obtained for each specification across the fixed, expanding-window, and rolling-window forecasting designs.

The lowest linear-model RMSFE is obtained by Model 2 under rolling-window estimation. The public-holiday specification and Model 1 perform similarly, with RMSFEs close to 159 cases. These results indicate that the joint covariate specifications provide more useful predictive information than most models containing only one additional predictor.

The AR(10) benchmark achieves an RMSFE of 199.76 cases and outperforms the AR(7) specification. Among the individual covariates, public-holiday information provides the largest reduction in forecast error.

Table 5: Forecast accuracy of the linear models measured by RMSFE on the original case scale

	RMSFE (Original Cases)		
Model specification	**Fixed test**	**Expanding**	**Rolling**
Linear model:Lagged cases only	200.12	202.24	202.57
Linear model + Public holiday	181.42	162.94	159.02
Linear model + Temperature	200.33	202.16	202.59
linear model + Reporting delay	199.14	201.66	202.28
Linear model + StartOfWeek	207.79	202.13	198.64
Linear model + Gender shares	200.84	200.89	212.90
Linear model + Age shares	200.90	231.60	206.51
Linear model + All covariates	171.03	164.31	158.55
Model 1	173.51	165.17	159.14
Model 2	171.50	164.75	158.47

Age-group and gender shares do not improve forecast accuracy when included separately. Rolling-window estimation performs best for the strongest linear specifications, suggesting that recent observations are particularly informative for their coefficients.

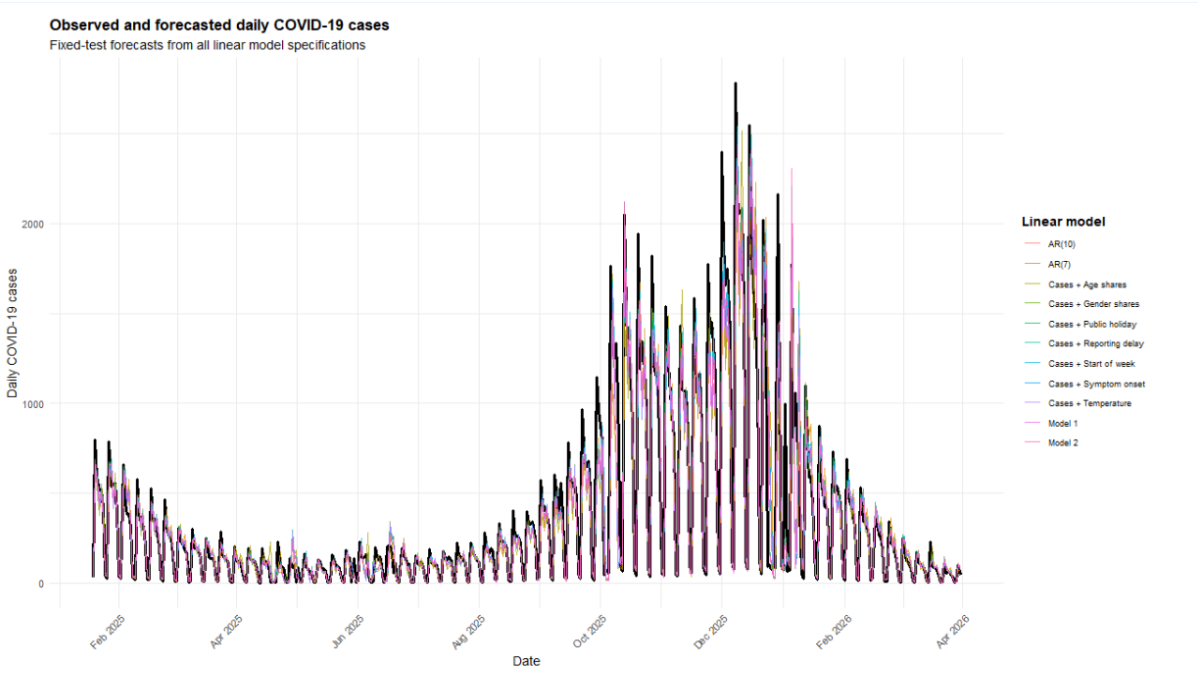


Figure 4: Observed and forecasted daily COVID-19 cases for all linear model specifications over the fixed test period.

Figure 4 compares the observed daily COVID-19 cases with the forecasts from all linear model specifications. All models follow the overall pattern of the observed data and capture the weekly reporting pattern, although larger forecast errors are observed during periods of rapid increases in cases. The forecasts from the different linear models are very similar, suggesting that adding individual exogenous variables provides only a small improvement over the benchmark autoregressive models.

4.6 Hyperparameter Selection

The regression tree and random forest models require several hyperparameters. These hyperparameters control the complexity of the models. They are selected using the validation sample by choosing the values that produce the lowest Root Mean Squared Forecast Error (RMSFE). Once selected, the same hyperparameter values are used to generate the final out-of-sample forecasts.

Table 6: Selected hyperparameters for the regression tree models

Model	**Min split**	**Max depth**	*cp*
Lagged cases only	20	5	0.001
Model 1	10	5	0.001
Model 2	10	5	0.001
All covariates	10	5	0.001

Table 6 shows the selected hyperparameters for the regression tree models. The maximum tree depth and the complexity parameter are the same for all model specifications. The only difference is the minimum split size. The lagged-cases model uses a minimum split of 20, while the other models use a value of 10. This suggests that similar hyperparameter values are suitable for all regression tree models.

Table 7: Selected hyperparameters for the random forest models

Model	**Number of trees**	*mtry*	**Minimum node size**
Lagged cases only	300	4	1
Model 1	100	20	5
Model 2	100	10	5
All covariates	200	20	1

Table 7 reports the selected hyperparameters for the random forest models. The selected values differ across the four model specifications. The lagged-cases model uses more trees and a smaller value of `mtry` because it contains fewer predictor variables. The other models use fewer trees and larger values of `mtry` to allow more predictor variables to be considered at each split. These results show that the optimal hyperparameter values depend on the number of predictor variables included in the model.

4.7 Regression Tree Models

Table 8 reports the forecasting performance of the regression tree models under the expanding-window and rolling-window forecasting designs.

Table 8: Forecast accuracy of the regression tree models measured by RMSFE on the original case scale

	RMSFE (Original Cases)	
Predictor specification	**Expanding**	**Rolling**
Lagged cases only	218.85	224.14
Model 1	207.62	203.54
Model 2	207.62	203.54
Lagged cases + All covariates	207.62	**203.54**

The lagged-cases model produces the largest forecast errors under both forecasting designs. The RMSFE is 218.85 cases for the expanding-window approach and 224.14 cases for the rolling-window approach. Adding the exogenous variables reduces the forecast error. All three augmented models produce the same RMSFE values. This indicates that adding more

predictor variables beyond Model 1 does not further improve the forecasting performance of the regression tree model. The rolling-window approach gives slightly lower forecast errors than the expanding-window approach for the augmented models.

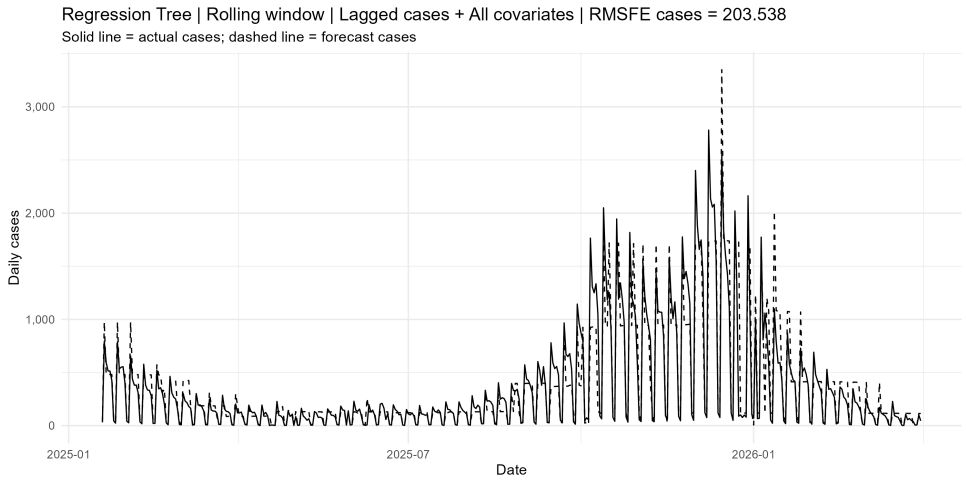


Figure 5: Actual vs Forecast plot for the best Regression trees model under the rolling-window forecasting design.

4.8 Random Forest Models

Table 9 presents the forecasting results of the random forest models for the expanding-window and rolling-window forecasting designs.

Table 9: Forecast accuracy of the random forest models

	RMSFE (Original Cases)	
Predictor specification	**Expanding**	**Rolling**
Lagged cases only	187.66	181.99
Model 1	142.42	**140.57**
Model 2	140.65	143.38
Lagged cases + All covariates	142.13	141.14

The lagged-cases model has the largest forecast error under both forecasting designs. The RMSFE is 187.66 cases for the expanding-window approach and 181.99 cases for the rolling-window approach. Adding the exogenous variables reduces the forecast error considerably for all model specifications.

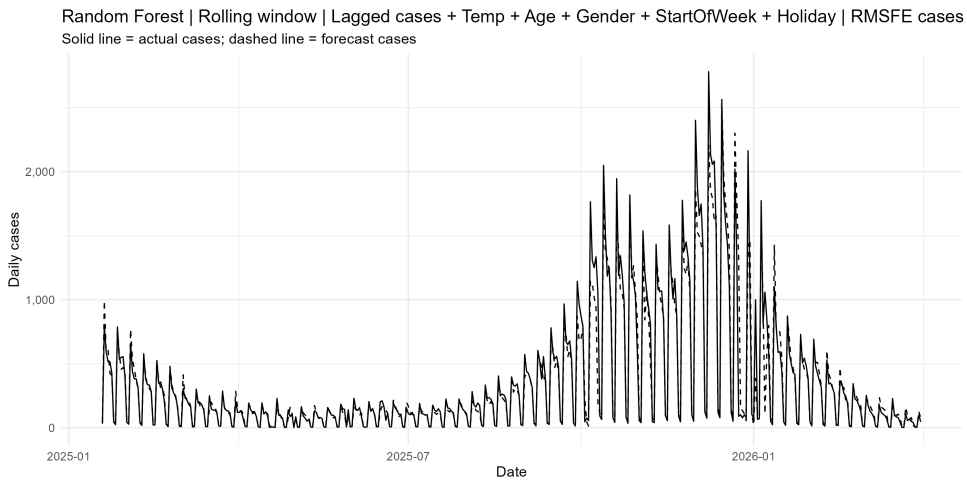


Figure 6: Actual vs Forecast plot for the best Random forest model under the rolling-window forecasting design.

The lowest forecast error is obtained by Model 1 under the rolling-window approach, with an RMSFE of 140.57 cases. Under the expanding-window approach, Model 2 performs slightly better, with an RMSFE of 140.65 cases. The model that includes all covariates also performs well, with RMSFE values of 142.13 and 141.14 cases for the expanding-window and rolling-window approaches, respectively.

4.9 Comparison of Forecast Accuracy

Table 10 compares the best-performing model from each forecasting method under the expanding-window and rolling-window forecasting designs. Forecast accuracy is measured using the Root Mean Squared Forecast Error (RMSFE). Lower RMSFE values indicate better forecasting performance.

Table 10: Best-performing specification from each primary model class

Model class	**Expanding window**	**Rolling window**
Random Forest	140.65	**140.57**
Linear ARX	163.70	158.47
Regression Tree	207.62	203.54

The random forest model gives the lowest RMSFE under both forecasting designs. The best result is obtained with the rolling-window approach, although the difference from the expanding-window approach is very small. This shows that the random forest model performs well under both forecasting designs.

The linear ARX model has the second-lowest forecast error. Its RMSFE decreases from 163.70 cases under the expanding-window approach to 158.47 cases under the rolling-window

approach. This suggests that using more recent observations improves the forecasting performance of the linear ARX model.

The regression tree model has the largest forecast errors under both forecasting designs. Although adding exogenous variables improves its performance compared with the lagged-cases model, it is still less accurate than the random forest and the linear ARX models.

4.10 Diebold–Mariano Tests

The Diebold–Mariano (DM) test is used to examine whether the differences in forecasting accuracy between two models are statistically significant. The test is based on the squared forecast errors over the common out-of-sample period. For the expanding-window forecasts, Random Forest Model 2 is used as the reference model because it has the lowest RMSFE. For the rolling-window forecasts, Random Forest Model 1 is used as the reference model.

Table 11: Best-performing models used as references in the Diebold–Mariano tests

Forecast design	**Reference model**	**RMSFE (Cases)**
Expanding window	Random Forest (Model 2)	140.65
Rolling window	Random Forest (Model 1)	140.57

Tables 12 and 13 report the results of the Diebold–Mariano tests for the expanding-window and rolling-window forecasting designs, respectively.

Table 12: Selected Diebold–Mariano tests for expanding-window forecasts

Comparison	**DM statistic**	***p*-value**
RF (Model 2) vs AR(10)	4.893	< 0.001
RF (Model 2) vs AR(7)	4.768	< 0.001
RF (Model 2) vs Linear ARX (Model 1)	2.435	0.015
RF (Model 2) vs Linear ARX (Model 2)	2.296	0.022
RF (Model 2) vs RF (Model 1)	4.351	< 0.001
RF (Model 2) vs RF (All covariates)	4.702	< 0.001

Table 13: Selected Diebold–Mariano tests for rolling-window forecasts

Comparison	**DM statistic**	***p*-value**
RF (Model 1) vs AR(10)	4.760	< 0.001
RF (Model 1) vs AR(7)	4.882	< 0.001
RF (Model 1) vs Linear ARX (Model 1)	2.406	0.016
RF (Model 1) vs Linear ARX (Model 2)	2.273	0.023
RF (Model 1) vs RF (Model 2)	0.480	0.632
RF (Model 1) vs RF (All covariates)	0.233	0.816

The results for the expanding-window forecasts show that Random Forest Model 2 performs significantly better than the AR(7), AR(10), and both linear ARX models. The p-values are all below 0.05, indicating that the differences in forecasting accuracy are statistically significant. Significant differences are also found when Random Forest Model 2 is compared with the other random forest models.

For the rolling-window forecasts, Random Forest Model 1 also performs significantly better than the AR(7), AR(10), and both linear ARX models. However, the differences between Random Forest Model 1, Random Forest Model 2, and the random forest model with all covariates are not statistically significant. These results suggest that the three covariate-augmented random forest models provide similar forecasting performance under the rolling-window forecasting design.

Harvey–Leybourne–Newbold (HLN) Correction

The Diebold–Mariano test results reported in this study use the Harvey–Leybourne–Newbold (HLN) correction. This correction improves the original Diebold–Mariano test when the number of forecast observations is small. Without the correction, the original test may show differences between models that are larger than they really are. The HLN correction provides more reliable p-values and makes the statistical results more accurate. Therefore, the model comparisons in this study are based on the HLN-corrected Diebold–Mariano test, which provides stronger evidence when comparing the forecasting performance of the different models. (Diebold & Mariano, 1995; D. Harvey et al., 1997).

5 Additional Forecasting Approach

The forecasting models presented in the previous sections focus on the short-term relationship between current and past COVID-19 cases. As an additional analysis, Seasonal Autoregressive Models with Exogenous Variables (SARX) are estimated to examine whether

including annual seasonal patterns improves forecast accuracy. The data cover almost six years, so observations from the same period in previous years may provide useful information for forecasting. Although the SARX model was not part of the original project, it is included as an additional analysis to examine whether seasonal patterns improve forecasting performance.

5.1 Seasonal Autoregressive Models with Exogenous Variables

Table 14 reports the forecasting performance of the SARX models under the expanding-window and rolling-window forecasting designs. The same four predictor specifications used in the previous models are also used for the SARX models.

Table 14: Forecast accuracy of the SARX models

	RMSFE (Original Cases)	
Predictor specification	**Expanding**	**Rolling**
Lagged cases only	157.10	157.10
Model 1	135.05	135.09
Model 2	135.58	135.61
Lagged cases + All covariates	**134.02**	134.06

The lagged-cases model has the largest forecast error under both forecasting designs. Adding the exogenous variables reduces the forecast error considerably. The best results are obtained when all covariates are included. The RMSFE is 134.02 cases under the expanding-window approach and 134.06 cases under the rolling-window approach.

The forecast errors are very similar for the expanding-window and rolling-window approaches. This suggests that the SARX model performs well under both forecasting designs.

5.2 Forecast Performance

Figures 7 and 8 compare the observed daily COVID-19 cases with the one-step-ahead forecasts from the best SARX model under the expanding-window and rolling-window approaches.

The predicted values follow the overall movement of the observed case numbers throughout the test period. Forecast errors are larger when case numbers change rapidly, but the model captures the main trend and the seasonal pattern in the data. These results are consistent with the low RMSFE values reported in Table 14.

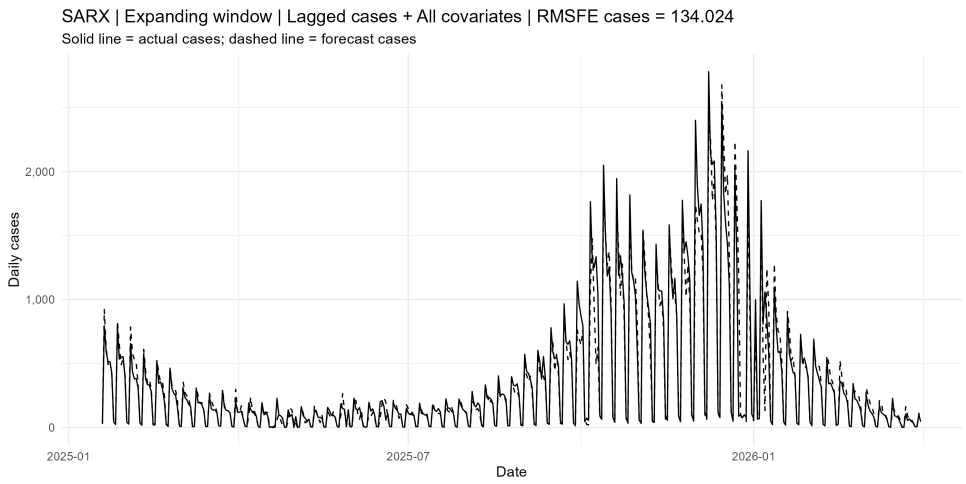


Figure 7: Observed daily COVID-19 cases and one-step-ahead forecasts from the best SARX model under the expanding-window forecasting design.

The predicted values follow the overall movement of the observed case numbers throughout the test period. Forecast errors are larger when case numbers change rapidly, but the model captures the main trend and the seasonal pattern in the data. These results are consistent with the low RMSFE values reported in Table 14.

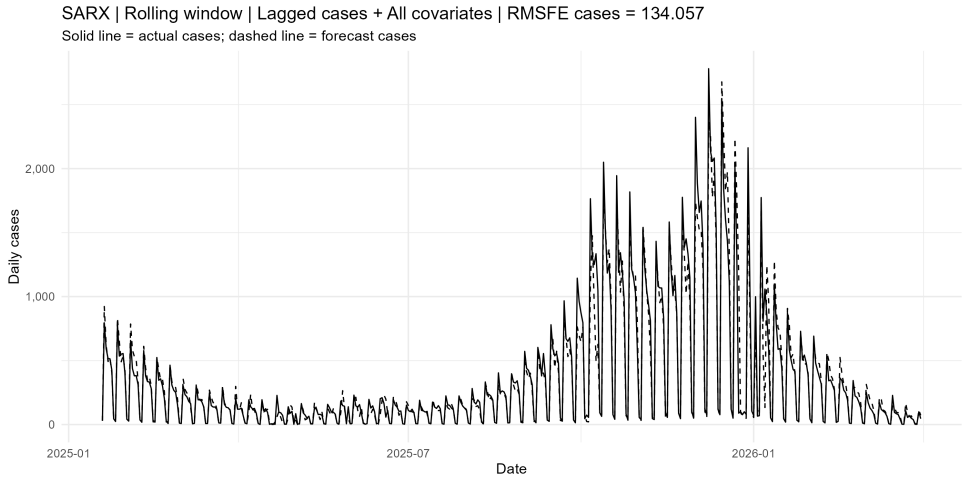


Figure 8: Observed daily COVID-19 cases and one-step-ahead forecasts from the best SARX model under the rolling-window forecasting design.

5.3 Comparison with the Main Forecasting Models

Table 15 compares the best SARX model with the best models from the main forecasting methods.

Table 15: Comparison of the best forecasting models

Forecasting method	**RMSFE (Cases)**
SARX	**134.02**
Random Forest	140.57
Linear ARX	158.47
Regression Tree	203.54

The SARX model has the lowest forecast error among all forecasting methods considered in this study. It performs better than the random forest, linear ARX, and regression tree models. These results show that including seasonal autoregressive terms improves forecasting accuracy for daily COVID-19 cases.

5.4 Relation to the Main Statistical Analysis

The Diebold–Mariano tests were carried out only for the forecasting methods included in the main analysis, namely the linear ARX models, regression trees, and random forests. Therefore, the SARX model was not included in these statistical comparisons.

The SARX results are evaluated using the RMSFE only. Although no formal statistical tests were performed, the SARX model produces the lowest forecast errors under both forecasting designs. This suggests that including annual seasonal information improves the one-step-ahead forecasts of daily COVID-19 cases.

6 Conclusion

This study compared the one-step-ahead forecasting performance of autoregressive models, linear autoregressive models with exogenous variables, regression trees, and random forests for daily reported COVID-19 cases in Germany. The models were evaluated using the same training, validation, and test periods under expanding-window and rolling-window forecasting designs. Forecast accuracy was measured using the Root Mean Squared Forecast Error (RMSFE).

The results show that including exogenous variables improves forecasting accuracy compared with models that use only lagged COVID-19 cases. Calendar and reporting-related variables are particularly useful because they capture regular patterns in the reporting process. Among the main forecasting methods, the random forest model produces the lowest forecast errors under both forecasting designs. The linear ARX models provide the second-best performance, while the regression tree models produce the largest forecast errors. The Diebold–Mariano tests show that the best random forest models perform significantly better than the autoregressive and linear benchmark models.

The results also show that the rolling-window approach performs slightly better than the expanding-window approach for the best linear ARX, regression tree, and random forest models. This suggests that recent observations provide more useful information for forecasting, as the relationship between COVID-19 cases and the predictor variables changes over time because of changes in testing, reporting practices, vaccination, and virus variants.

As an additional analysis, the Seasonal Autoregressive Model with Exogenous Variables (SARX) achieves the lowest RMSFE in the study. This suggests that annual seasonal patterns contain useful information for forecasting daily COVID-19 cases beyond short-term dependence and the included predictor variables. Since the SARX model was introduced as an additional analysis and was not included in the Diebold–Mariano tests or the Model Confidence Set, its results should be interpreted as additional evidence rather than as part of the formal comparison of the primary forecasting models.

This study has some limitations. The analysis is based on reported national COVID-19 cases, which are affected by testing behaviour, reporting delays, data revisions, and administrative practices. In addition, only one-step-ahead forecasts are considered, and the analysis focuses on point forecasts rather than prediction intervals.

Future research could extend this work by considering regional forecasts, longer forecasting horizons, probabilistic forecasting methods, and additional machine-learning models. Combining seasonal time-series models with ensemble learning methods may also provide further improvements in forecasting performance.

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A Variable Description

Table 16: Description of Variables in the RKI COVID-19 Dataset

Variable	**Description**
IdLandkreis	Unique identifier of the German district reporting the observation.
Landkreis	Name of the reporting district.
Meldedatum	Date on which the case or death was reported to the RKI.
Refdatum	Reference date, usually symptom onset date if available.
Altersgruppe	Age group of the reported individual.
Geschlecht	Gender of the reported individual.
AnzahlFall	Number of reported COVID-19 cases in the observation.
AnzahlTodesfall	Number of reported COVID-19 deaths in the observation.
AnzahlGenesen	Number of reported recoveries in the observation.
IstErkrankungsbeginn	Indicator equal to 1 if the reference date corresponds to symptom onset.
NeuerFall	Indicator showing whether a case record is new, updated, or corrected.
NeuerTodesfall	Indicator showing whether a death record is new, updated, or corrected.
NeuGenesen	Indicator showing whether a recovery record is new, updated, or corrected.